Branch: MCA (Data Science) Kargil	**Semester:** 2
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Subject Name: Technical Training - I Lab	**Subject Code**: 25CAP-652
Section/Group: 25MCD KAR-1	**Date of Performance:** 31-03-2026

Experiment 7

Experiment Aim:

Implementation of Joins in PostgreSQL to retrieve and combine data from multiple related tables. (Company Tags: Oracle, SAP, IBM, Microsoft)

Tools used: PostgreSQL

Objectives:

- **Join Operations:** To understand how different types of joins (INNER, LEFT, RIGHT, CROSS) are used to combine data from multiple relations.
- **Data Retrieval:** To learn how to extract meaningful information from relational schemas using join conditions.
- **Handling Missing Data:** To identify and manage NULL values in join operations.
- **Query Optimization:** To write efficient queries for real-world database scenarios.

Theory:

A JOIN in SQL is used to combine rows from two or more tables based on a related column between them. Joins are fundamental in relational databases as they allow retrieval of distributed data efficiently.

1. INNER JOIN: Returns only matching records from both tables.
2. LEFT JOIN: Returns all records from the left table and matching records from the right table.
3. RIGHT JOIN: Returns all records from the right table and matching records from the left table.
4. CROSS JOIN: Returns the Cartesian product of both tables.
5. MULTIPLE JOIN: Combines more than two tables using join conditions.

Experiment Steps:

Step 1: Creating Base Tables

Query:

```
-- Create Students table
CREATE TABLE Students (
student_id INT PRIMARY KEY,
name VARCHAR(50),
dept_id INT );
```

```
-- Create Courses table
CREATE TABLE Courses (
course_id INT PRIMARY KEY,
course_name VARCHAR(50)
);

-- Create Enrollments table
CREATE TABLE Enrollments (
student_id INT,
course_id INT,
PRIMARY KEY (student_id, course_id)
);

-- Create Departments table
CREATE TABLE Departments (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50)
);
```

Step 2: Inserting Sample Data

Query:

```
-- Students
```

```
INSERT INTO Students VALUES
(1, 'Rahul', 101),
(2, 'Priya', 102),
(3, 'Amit', 101),
(4, 'Neha', 103);

-- Courses
INSERT INTO Courses VALUES
(201, 'DBMS'),
(202, 'AI'),
(203, 'ML');

-- Enrollments
INSERT INTO Enrollments VALUES
(1, 201),
(1, 202),
(2, 202),
(3, 203);

-- Departments
INSERT INTO Departments VALUES
(101, 'CSE'),
(102, 'IT'),
(103, 'ECE');
```

Step 3: INNER JOIN (Students with Enrolled Courses)

Query:

```
SELECT s.name AS student_name, c.course_name
FROM Students s
INNER JOIN Enrollments e
ON s.student_id = e.student_id
INNER JOIN Courses c
ON e.course_id = c.course_id;
```

Output:

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

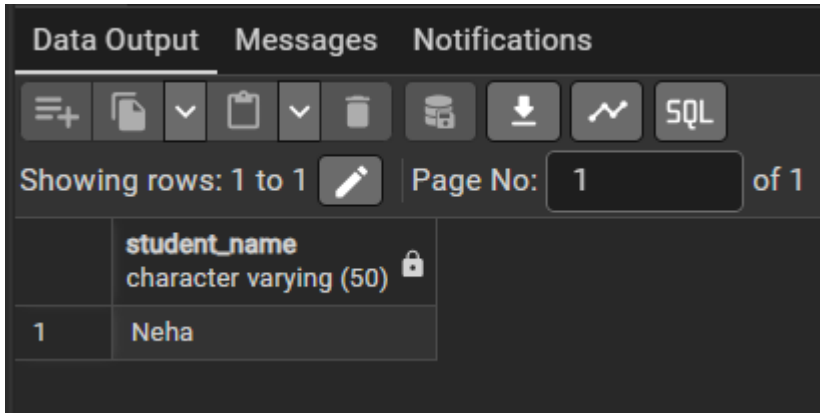
	student_name character varying (50)	course_name character varying (50)
1	Rahul	DBMS
2	Rahul	AI
3	Priya	AI
4	Amit	ML

Step 4: LEFT JOIN (Students not enrolled in any course)

Query:

```
SELECT s.name AS student_name
FROM Students s
LEFT JOIN Enrollments e
ON s.student_id = e.student_id
WHERE e.course_id IS NULL;
```

Output:



Step 5: RIGHT JOIN (All courses with or without students)

Query:

```
SELECT c.course_name, s.name AS student_name
```

```
FROM Students s
RIGHT JOIN Enrollments e
ON s.student_id = e.student_id
RIGHT JOIN Courses c
ON e.course_id = c.course_id;
```

Output:

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	course_name character varying (50)	student_name character varying (50)
1	DBMS	Rahul
2	AI	Rahul
3	AI	Priya
4	ML	Amit

Step 6: MULTIPLE JOIN (Students with Department Info)

Query:

```
SELECT s.name AS student_name, d.dept_name
FROM Students s
```

```
JOIN Departments d
ON s.dept_id = d.dept_id;
```

Output:

Showing rows: 1 to 4 Page No: 1 of 1

	student_name character varying (50)	dept_name character varying (50)
1	Rahul	CSE
2	Priya	IT
3	Amit	CSE
4	Neha	ECE

Step 7: CROSS JOIN (All possible combinations)

Query:

```
SELECT s.name AS student_name, c.course_name
FROM Students s
CROSS JOIN Courses c;
```

Output:

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

	student_name character varying (50)	course_name character varying (50)
1	Rahul	DBMS
2	Priya	DBMS
3	Amit	DBMS
4	Neha	DBMS
5	Rahul	AI
6	Priya	AI
7	Amit	AI
8	Neha	AI
9	Rahul	ML
10	Priya	ML
11	Amit	ML
12	Neha	ML

Outcomes:

- Students will be able to implement different types of joins in SQL.
- Students will understand how to retrieve and combine related data from multiple tables.

- Students will gain hands-on experience in handling NULL values and join conditions.
- Students will be able to apply join operations in real-world database scenarios.